

AKSHUL JAMWAL

☎ +91-7018538655 ✉ akshuljamwal@gmail.com in Akshul jamwal 🌐 Akshul1

EDUCATION

Bahra University

Bachelor of Technology in Computer Science - **CGPA - 7.01**

August 2022 – June 2026

Wakhnaghat, Himachal Pradesh

Govt.(boys)Sen.Sec.School

Intermediate - **Percentage - 69.9%**

March 2022

Nadaun, Himachal Pradesh

Kendriya Vidyalaya

Secondary - **Percentage - 75.6%**

March 2020

Nadaun, Himachal Pradesh

PROJECTS

Movie Reservation System 📄 | Spring Boot, PostgreSQL, JWT, Docker

Oct 2025 – Present

- Engineered a scalable RESTful API for cinema ticket management, implementing **Concurrency Control** through a custom Seat-Locking manager to prevent double-booking in high-traffic scenarios.
- Developed a robust security layer using **Spring Security and JWT** (JSON Web Tokens), enabling stateless authentication and fine-grained Role-Based Access Control (RBAC) for Admin and User roles.
- Architected a relational database schema in **PostgreSQL**, utilizing Spring Data JPA and Hibernate for efficient data persistence and complex relationship mapping between Movies, Shows, and Seats.
- Optimized the development workflow by **Containerizing** the application and database using **Docker**, ensuring consistent environments across development and testing phases.
- Integrated custom exception handling and global error responses, improving the API's reliability and providing a seamless integration experience for potential frontend consumers.

Cube - Breaker 📄 | Java, Java Swing, OpenCV (Computer Vision), Git, GitHub

July 2025

- Implemented a **Computer Vision** pipeline using OpenCV to perform real-time color detection and edge processing, allowing accurate scanning of Rubik's Cube faces via webcam.
- Developed an automated solver using the **Kociemba Algorithm** (or Layer-by-Layer logic) to compute the mathematically optimal solution in under 20 moves from any scrambled state.
- Engineered a **State Management System** using multi-dimensional arrays and matrix transformations to represent cube rotations and track the orientation of 26 individual cubies.
- Designed a custom GUI with Java Swing that features a 2D-unfolded cube visualization, allowing users to verify scan accuracy before executing the step-by-step solving logic.

Personal Portfolio & Professional Showreel 📄 | HTML, CSS, JavaScript, Git, GitHub

March 2025

- Architected a high-performance personal portfolio using **semantic HTML** and modular CSS to showcase technical engineering projects and professional background.
- Engineered a **mobile-first responsive interface** utilizing CSS Flexbox and Grid, ensuring a seamless user experience across a wide range of devices and screen resolutions.
- Optimized web performance and SEO through efficient asset management and clean DOM structuring, resulting in fast load times and improved searchability.
- Deployed and maintained the site via **GitHub Pages**, integrating a custom domain and using Git for version control and continuous updates.

TECHNICAL SKILLS

Languages: Java, SQL, HTML, CSS.

Frameworks & Libraries: Spring Boot, Spring MVC, Spring Security, Spring Data JPA, Hibernate, JUnit .

Developer Tools: IntelliJ IDEA, VS Code, Postman, Android Studio, Git, GitHub.

Core Fundamentals: Object-Oriented Programming (OOPS), Database Management Systems (DBMS), RESTful API Design.

DevOps: Docker (Containerization), Git (Version Control)

EXTRACURRICULAR & ACHIEVEMENTS

- * **Version Control:** Maintaining projects on GitHub with clean commit histories.
- * **Technical Upskilling:** Self-taught Spring Boot and Docker for backend development.
- * **Academic Collaboration:** Engaged in peer code reviews to ensure OOPS principles.

CERTIFICATIONS

• Core Java - MCP TECHNOLOGY

• Java